CNC PROGRAM SHEET

Part: Motor Mount | Rev: A | Date: 2026-06-24

Programmer: ShiaanX-AI

Material: ALUMINIUM

Max RPM: 4500 Coolant: through spindle

Setups: 3

Part envelope: 55.3 x 104.8 x 21.5 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face center of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	CD_60_2MM	60-degree combined center drill, 2mm tip, 5mm body	2.0	Sandvik Coromant	H10F	spot_drill
T02	EM_2FL_2MM	2-flute solid carbide end mill 2mm â€” aluminium geometry	2.0	Sandvik Coromant	1630	contour_mill
T03	DRILL_2.5MM	Solid carbide jobber drill 2.5mm	2.5	Sandvik Coromant	H10F	twist_drill
T04	DRILL_3.0MM	Solid carbide jobber drill 3.0mm	3.0	Sandvik Coromant	H10F	twist_drill
T05	DRILL_3.2MM	Solid carbide jobber drill 3.2mm â€” standard size for Ã˜3.2mm pin holes	3.2	Sandvik Coromant	H10F	twist_drill
T06	DRILL_4.0MM	Solid carbide jobber drill 4.0mm	4.0	Sandvik Coromant	H10F	twist_drill
T07	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T08	BALL_2FL_AL_4MM	2-flute solid carbide ball-nose end mill 4mm â€” aluminium geometry	4.0	Sandvik Coromant	1630	profile_3d_contour
T09	DRILL_5.0MM	Solid carbide jobber drill 5.0mm	5.0	Sandvik Coromant	H10F	twist_drill
T10	SPOT_090_6MM	90-degree solid carbide spot drill, 6mm shank	6.0	Sandvik Coromant	H10F	spot_drill
T11	EM_2FL_6MM	2-flute solid carbide end mill 6mm â€” aluminium geometry	6.0	Sandvik Coromant	1630	pocket_mill
T12	CHAM_90_6MM	90-degree solid carbide chamfer mill 6mm, 2-flute	6.0	Sandvik Coromant	1630	chamfer_mill
T13	DRILL_10.0MM	Solid carbide jobber drill 10.0mm â€” also used as pilot drill	10.0	Sandvik Coromant	H10F	pilot_drill, twist_drill
T14	DRILL_16.0MM	Solid carbide jobber drill 16.0mm â€” also used as core drill	16.0	Sandvik Coromant	H10F	core_drill
T15	EM_2FL_20MM	2-flute solid carbide end mill 20mm â€” aluminium geometry	20.0	Sandvik Coromant	1630	circular_interp
T16	FACEMILL_40MM	40mm indexable face mill â€” 6 inserts	40.0	Sandvik Coromant		face_mill
T17	FACEMILL_50MM	50mm indexable face mill / shell mill â€” 4 inserts	50.0	Sandvik Coromant		face_mill

SETUP 1 -- -Z approach (rear face)

PRINCIPAL | 42 clusters | 72 operations

Fixture	90° angle plate or tombstone fixture.
Description	Rear-face setup. Rotate part 90° so rear face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=0.000mm) Y=CENTER (Y=-20.711mm) Z=TOP
Face (W x H)	55.3 x 104.8 mm
Max depth	21.5 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
2	ALU 4 DRILL	T06	4.0	0	0.41	1590	95	0.408	--	--	00:01
3	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
4	ALU 4 DRILL	T06	4.0	0	0.41	1590	95	0.408	--	--	00:01
5	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
6	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
7	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
8	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
9	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
10	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
11	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
12	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
13	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
14	ALU 4 DRILL	T06	4.0	0	0.33	1590	95	0.332	--	--	00:01
15	ALU 50 FACEMILL BORE FINISH	T17	50.0	0	0.29	4500	2700	0.294	0.0	0.0	00:01
16	ALU 20 ENDMILL BORE	T15	20.0	0	0.29	4500	540	0.294	--	--	00:10
17	ALU 50 FACEMILL BORE FINISH	T17	50.0	0	0.29	4500	2700	0.294	0.0	0.0	00:01
18	ALU 20 ENDMILL BORE	T15	20.0	0	0.29	4500	540	0.294	--	--	00:10
19	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
20	ALU 4 DRILL	T06	4.0	0	0.33	1590	95	0.332	--	--	00:01
21	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
22	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
23	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
24	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
25	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
26	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
27	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
29	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
30	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
31	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
32	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
33	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
34	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
35	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
36	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
37	ALU CENTER DRILL	T10	6.0	0	3.0	420	21	3.0	--	--	00:09
38	ALU 10 DRILL	T13	10.0	0	1.5	4500	495	1.5	--	--	00:01
39	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
40	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
41	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
42	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
43	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
44	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
45	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
46	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
47	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
48	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
49	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
50	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
51	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
52	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
53	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
54	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
55	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
56	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
57	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.01	4500	90	0.009	0.1	0.1	00:06
58	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.01	4500	90	0.009	0.0	0.0	00:06
59	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.02	4500	90	0.019	0.1	0.1	00:06
60	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.02	4500	90	0.019	0.0	0.0	00:06
61	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.03	3580	358	0.025	0.0	0.0	00:10
62	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
63	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
64	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:10
65	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:10
66	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
67	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T8	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
68	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T8	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:10
69	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T8	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:10
70	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T8	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:10
71	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T8	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:10
72	ALU 40 FACEMILL FACE FINISH	T16	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05

ESTIMATED SETUP TIME: 05:45 (includes tool changes)

SETUP 2 -- +Y approach (top — default VMC position)

PRINCIPAL | 16 clusters | 23 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (X=0.000mm, 27.67mm from xmin edge) Y=CENTER (Z=2.750mm, 10.76mm from zmin edge) Z=TOP
Face (W x H)	55.3 x 21.5 mm
Max depth	104.8 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
2	ALU 5 DRILL	T09	5.0	0	9.99	1400	98	9.988	--	--	00:06
3	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
4	ALU 5 DRILL	T09	5.0	0	9.4	1400	98	9.405	--	--	00:06
5	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
6	ALU 5 DRILL	T09	5.0	0	9.4	1400	98	9.405	--	--	00:06
7	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
8	ALU 5 DRILL	T09	5.0	0	9.99	1400	98	9.988	--	--	00:06
9	ALU CENTER DRILL	T10	6.0	0	3.0	420	21	3.0	--	--	00:09
10	ALU 10 DRILL	T13	10.0	0	17.03	4500	540	17.033	--	--	00:02
11	ALU 16 DRILL	T14	16.0	0	17.03	3580	573	17.033	--	--	00:02
12	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	2.54	3580	358	0.4	0.0	0.0	00:10
13	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.05	3580	358	0.051	0.0	0.0	00:10
14	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	4.1	3580	358	0.4	0.0	0.0	00:10
15	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	2.37	3580	358	0.4	0.0	0.0	00:10
16	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	4.27	3580	358	0.4	0.0	0.0	00:10
17	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.05	3580	358	0.051	0.0	0.0	00:10
18	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
19	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
20	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
21	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
22	ALU 6 ENDMILL POCKET RF	T11	6.0	0	--	3020	483	6.0	0.1	0.1	00:05
23	ALU 6 ENDMILL POCKET FINISH	T11	6.0	0	--	3020	483	6.0	0.0	0.0	00:05

ESTIMATED SETUP TIME: 03:44 (includes tool changes)

SETUP 3 -- -X approach (left side face)

PRINCIPAL | 12 clusters | 14 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (Y=-20.711mm) Y=CENTER (Z=2.750mm) Z=TOP
Face (W x H)	104.8 x 21.5 mm
Max depth	55.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
2	ALU 5 DRILL	T09	5.0	0	0.74	1400	98	0.738	--	--	00:01
3	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
4	ALU 3 DRILL	T04	3.0	0	3.57	1170	78	3.573	--	--	00:03
5	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
6	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
7	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
8	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
9	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	3.0	3580	358	0.4	0.0	0.0	00:10
10	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	4.75	3580	358	0.4	0.0	0.0	00:10
11	ALU 4.0 PROFILE 3D CONTOUR FILLET FINISH	T08	4.0	0	1.69	3580	358	0.4	0.0	0.0	00:10
12	ALU 40 FACEMILL FACE FINISH	T16	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
13	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	4500	540	2.0	--	--	00:10
14	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	4500	540	2.0	--	--	00:10

ESTIMATED SETUP TIME: 02:40 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C3	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C7	face_mill	Effective Vc = 706.9 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	circular_interp	Effective Vc = 282.7 m/min (reduced from 500 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	face_mill	Effective Vc = 706.9 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	circular_interp	Effective Vc = 282.7 m/min (reduced from 500 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C18	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C19	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C20	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C21	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
WARNING	C22	spot_drill	WARNING: no spot drill covers dia=16.1mm, using largest available 6.0mm Field-validated: RPM ~425 F21 â€” slow spot for p
SUBSTITUTION	C22	pilot_drill	SUBSTITUTION: 9.66mm is non-standard. Using nearest standard size 10.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C22	pilot_drill	Effective Vc = 141.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C22	core_drill	SUBSTITUTION: 16.1mm is non-standard. Using nearest standard size 16.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C25	twist_drill	Effective Vc = 141.4 m/min (reduced from 220 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C31	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C32	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C33	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C34	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C65	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C67	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C68	chamfer_mill	Effective Vc = 84.8 m/min (reduced from 85 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C69	chamfer_mill	Effective Vc = 84.8 m/min (reduced from 85 m/min). Feed rate adjusted accordingly.

SETUP 1 — FIXTURE SPECIFICATION WCS: G54 | -Z approach (rear face)

Field	Value
Fixture Type	Angle Plate
Jaw Type	Standard Hard Jaws
Jaw Opening	70 mm
Jaw Height	38 mm
Parallels	10 mm
Clamped Faces	-Y, +X
Rest Face	-Y
Datum Face	+Z
Z Zero Reference	Top Of Stock
WCS XY Zero	Center Of Part
Clamp Clearance	1.2 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off top face of raw billet
- Use 10mm parallels under part â€” seat firmly before torquing vise
- Minimum 7mm jaw engagement each side
- Angle plate setup â€” verify squareness to spindle before zeroing

SETUP 2 — FIXTURE SPECIFICATION WCS: G55 | +Y approach (top â€” default VMC position)

Field	Value
Fixture Type	Fixture Plate
Jaw Type	Standard Hard Jaws
Jaw Opening	70 mm
Jaw Height	38 mm
Parallels	10 mm
Clamped Faces	+X, -X
Rest Face	-Y
Datum Face	-Y
Z Zero Reference	Top Of Part
WCS XY Zero	Center Of Part
Clamp Clearance	9.5 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off previously machined face (probe before clamping)
- Use 10mm parallels under part â€” seat firmly before torquing vise
- Minimum 7mm jaw engagement each side
- Fixture plate â€ locate on dowel pins, clamp with M8 bolts
- Datum from Setup 1 â€ use finished face as reference

SETUP 3 — FIXTURE SPECIFICATION WCS: G56 | -X approach (left side face)

Field	Value
Fixture Type	Angle Plate
Jaw Type	Standard Hard Jaws
Jaw Opening	35 mm
Jaw Height	38 mm
Parallels	None (part sits on jaw serrations)
Clamped Faces	-Y, +Z
Rest Face	-Y
Datum Face	+X
Z Zero Reference	Top Of Part
WCS XY Zero	Center Of Part
Clamp Clearance	5.5 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off previously machined face (probe before clamping)
- Jaw opening set to 35mm â€” verify engagement before cutting
- Angle plate setup â€” verify squareness to spindle before zeroing
- Datum from Setup 2 â€” use finished face as reference